

A DATA-AWARE DEEP SUPERVISED METHOD FOR RETINAL VESSEL SEGMENTATION

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ABSTRACT

Accurate vessel segmentation in retinal images is vital for retinopathy diagnosis and analysis. However, very thin vessels in low image contrast along with pathological conditions (e.g., capillary dilation or microaneurysms) render the segmentation task difficult. In this work, we present a novel approach for retinal vessel segmentation focusing on improving thin vessel segmentation. We develop a deep convolutional neural network (CNN), which exploits the specific characteristics of the input retinal data to use deep supervision, for improved segmentation accuracy. In particular, we use the average input retinal vessel width and match it with the layer-wise effective receptive fields (*LERF*) of the CNN to determine the location of the auxiliary supervision. This helps the network to pay more attention to thin vessels, that otherwise the network would ‘ignore’ during training. We verify our method on three public retinal vessel segmentation datasets (DRIVE, CHASE_DB1, and STARE), achieving better sensitivity (10.18% average increase) than state-of-the-art methods while maintaining comparable specificity, accuracy, and AUC.

Index Terms— Retinal Vessel Segmentation, Effective Receptive Field, Convolutional Neural Networks

1. INTRODUCTION

Analyzing retinal vessels helps the diagnosis and treatment of various ophthalmological diseases. An example of such major ocular problems is diabetic retinopathy, for which retinal vascular structures are critical for inspection and screening. Hence, accurate vessel segmentation in retinal images is vital for early retinopathy diagnosis. However, conditions like local intensity change, vessel resolution, low image contrast, background noise, and presence of lesions, render this segmentation task challenging. This situation exacerbates with extremely thin vessels (with a width of a few pixels).

Considerable work has been done on retinal vessel segmentation. [1] was a significant earlier work that used hand-craft features in segmentation. In [2], vessel segmentation was viewed as a cross modality data transformation problem from a retinal image to a vessel map. Discriminatively trained

fully connected conditional random fields were explored in [3]. Introduction of additional labels are proposed in [4], for better boundary detection. [5] proposed a multi-scale, network followed network model, for enhanced feature extraction. Despite their success, most of these methods fail to segment thin vessels accurately, resulting in lower sensitivity compared to other accuracy metrics. Giving high priority to segmenting thin vessels by adjusting the networks’ internal feature extraction mechanics is a main objective of this work.

To achieve the objective of thin vessel segmentation, ensuring the discriminativeness of the learned features in the early network layers is important, owing to the small widths of the vessels. Deep supervision using auxiliary classifiers [6] is efficient in encouraging the feature maps at earlier layers to be directly predictive of the final labels. Deep supervision can be viewed as a feedback generated by a local output (from a hidden layer) by providing companion objective with auxiliary classifier [7]. During back-propagation, the error proceeds simultaneously from the local (auxiliary) and the global (main) output, favoring highly discriminative feature-maps. However, the questions of where and how to add deep supervision were not addressed in [6]. [7] extended this idea by proposing all intermediate layers to contribute to their individual auxiliary classifiers. [4] proposed random auxiliary classifiers for retinal vessel segmentation. We extend the idea of deep supervision to retinal vessel segmentation by proposing data-aware deep supervision. For the data-aware part, using the training data, we determine the layer which has the preeminent contribution to vessel feature extraction. For the deep supervision part, the preeminent layer is used as the contributor for auxiliary classification. We employ a receptive field based approach to determine the preeminent layer.

In CNNs, the output of a convolutional layer is connected only to a region of its input. This region affects the output’s values and is known as the layer’s *receptive field* (*RF*) [8]. For a CNN performing segmentation, *RF* of the network is defined as the region of the input affecting the output of the encoder. [9] showed that this output-affecting region is even smaller than the theoretical *RF* and called it the *effective receptive field* (*ERF*) of the network. Intuitively, convolutional filters extract improved features when the *target object size* matches the filters’ *ERF* and lies at the center of *ERF*.

From the view point of *ERF* matching, for retinal vessel segmentation using a deep CNN, there are two challenges: (1) determining the *target object size*, i.e., the characteristic size of vessels for the whole dataset; (2) determining the network *ERF*, which has to be matched with the *target object size*. For the first challenge, we use the average local vessel width as the characteristic vessel size. For the second challenge, we analyze the network's *ERF* and propose a new variation of it. In retinal images, thin vessels often have a width of only a few pixels, and are "lost" in the deeper layers of the CNN after a certain number of sub-sampling operations. Hence, instead of using the relatively large network *ERF* of a deep CNN, for matching with the thin retinal vessels, we modify the definition of *ERF*. We propose the layer-wise *ERF*, *LERF*, which represents *ERF* of a convolutional layer in the network with respect to the network's input. The encoder layer with *LERF* similar to characteristic vessel size, is considered as the preeminent layer and is used for deep supervision.

In data-aware deep supervision, we first compute the average vessel width of the labeled training data, which we take as the vessel size. Then we introduce deep supervision from the layer of the network whose *LERF* matches (as well as possible) the vessel size. Data-aware deep supervision encourages the network to pay extra attention to thin vessels and not to ignore them as with the presence of the auxiliary classifier they contribute directly in loss calculation. Further, we propose a new network architecture combining the decoders of U-Net [10] and CUMed-Vision [11] to assist vessel segmentation by facilitating contextual information flow. We validate our method on 3 public retinal vessel segmentation datasets (DRIVE, CHASE.DB1, and STARE), and show that our method outperforms state-of-the-art methods in sensitivity by an average of 10.18%, while maintaining comparable specificity, accuracy, and AUC.

2. METHODOLOGY

Our proposed framework is shown in Fig. 1. The major components of our framework are: (1) *LERF* determination, (2) vessel width calculation, and (3) data-aware deep supervision. Section 2.1 presents *LERF* determination, and Section 2.2 highlights vessel width calculation. We then show how to use these components for data-aware deep supervision. Finally, we elaborate our proposed network.

2.1. LERF Determination

Consider a CNN with N convolutional layers in its encoder. For each layer h , *LERF* quantifies the region of influence of the input over a output node (a unit feature map, i.e., a pixel) of the specific layer h . Influence measurement is performed using partial derivative, showing the relative change of output over the change in input [9]. Since back-propagation effectively computes error gradient with respect to a loss func-

tion (*Loss*), i.e., $\frac{\partial \text{Loss}}{\partial x_{i,j}}$, applying the chain rule and assigning $\frac{\partial \text{Loss}}{\partial y_{i_t,j_t}} = 1$ and $\frac{\partial \text{Loss}}{\partial y_{i,j}} = 0$, for $i_t \neq i$ and $j_t \neq j$, we can compute $\frac{\partial y_{i_t,j_t}}{\partial x_{i,j}}$, where $x_{i,j}$ is a node in the input layer with indices (i, j) and y_{i_t,j_t} is the target node in the output layer h on which the *LERF* calculation is being performed. $\frac{\partial y_{i_t,j_t}}{\partial x_{i,j}}$ indicates the region of input (i.e., all $x_{i,j}$'s) influencing the output node (i.e., y_{i_t,j_t}). The algorithm for computing *LERF* using this bottom-up approach is shown in Algorithm 1. Examining the near-Gaussian distribution of *ERF* [9], the square root of the number of nonzero nodes in the thresholded $\frac{\partial y_{i_t,j_t}}{\partial x_{i,j}}$ is considered as *LERF* to ignore regions with non-significant contribution.

Algorithm 1: *LERF* Determination

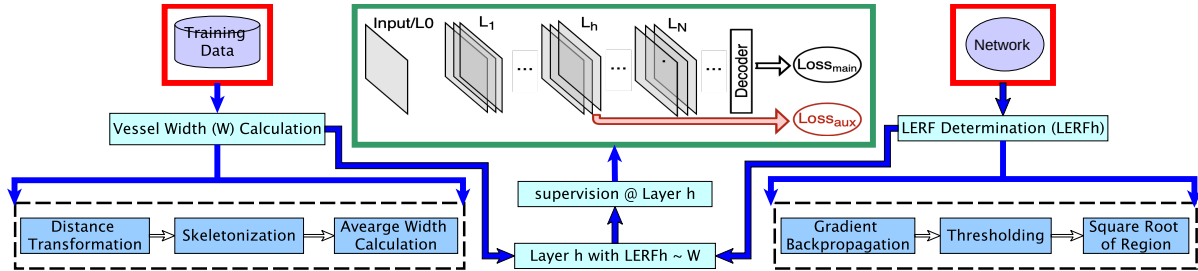
input : A CNN architecture with N encoder layers.
output: $LERF_h, \forall h \in N$.
for $h \in N$ **do**
 select a node y_{i_t,j_t} for conv-layer h ;
 $\frac{\partial \text{Loss}}{\partial y_{i_t,j_t}} = 1$ and $\frac{\partial \text{Loss}}{\partial y_{i,j}} = 0$ for $i_t \neq i$ and $j_t \neq j$, where (i, j) is a feature-map node;
 compute $\frac{\partial \text{Loss}}{\partial x_{i,j}}$ using back-propagation;
 compute $\frac{\partial y_{i_t,j_t}}{\partial x_{i,j}}$ as $\frac{\partial \text{Loss}}{\partial x_{i,j}} = \sum_{i,j} \frac{\partial \text{Loss}}{\partial y_{i_t,j_t}} \frac{\partial y_{i_t,j_t}}{\partial x_{i,j}}$, where $x_{i,j}$ is an arbitrary node in the input layer with indices (i, j) ;
 $z_{i,j} = \frac{\partial y_{i_t,j_t}}{\partial x_{i,j}} > \text{threshold}$;
 $LERF_h = \sqrt{|z_{i,j} \neq 0|}$
end
return $LERF_h \forall h$

2.2. Vessel Width Calculation

We use a skeletonization [12] based approach for vessel width calculation on the training set of labelled images $M_i, i = 1, 2, \dots, m$. The distance transformed labelled images (D_i) are mapped to the skeleton (medial-axis) of vessels in M_i . This generates the halved vessel width at each point of the skeleton. The mean value of twice of these halved width values is considered as the average vessel width for M_i . The dataset characteristic vessel width is the mean value taken over all the m labels. The algorithm used for vessel width calculation is shown in Algorithm 2.

Algorithm 2: Vessel Width Calculation

input : Training labels: $M = \{M_1, M_2, \dots, M_m\}$.
output: Average vessel width W .
for $M_i \in M$ **do**
 $C_i = \text{complement}(M_i)$;
 $D_i = \text{distance-transform}(C_i)$;
 $S_i = \text{skeletonize}(M_i)$;
 $\text{temp}_i = (D_i \cdot S_i \neq 0)$;
 $\text{Width}_{avg_i} = 2 \times \text{mean}(\text{temp}_i)$
end
return $W = \text{mean}(\text{Width}_{avg_i})$



2.3. Data-Aware Deep Supervision

Using $LERF$ computed in Sec. 2.1 and the vessel width (W) calculated in Sec. 2.2, we determine the convolutional layer (L_{target}) with preeminent contribution towards vessel feature extraction. The convolutional layer L_h with the minimum absolute difference between $LERF_h$ and W is selected as L_{target} . Since our objective is to include the features extracted by L_{target} into network loss calculation, we introduce auxiliary loss from L_{target} . To perform deep supervision, we need to up-sample the sub-sampled feature-maps generated by L_{target} which will incur additional network weights. To restrict these weights, we introduce deep supervision from the output in the decoder part of the model which has a direct connection with L_{target} and has already up-sampled the feature-maps. If no such direct connection exists, then the next nearest conv-layer to L_{target} , L_{h^*} , is selected, with $h^* > target$.

Our network has a main branch loss $Loss_{main}$, and the auxiliary loss from data-aware deep supervision, $Loss_{aux}$. The total loss of the network is defined as $Loss_{total} = Loss_{main} + Loss_{aux} + L_2\text{-Regularization}$, where both $Loss_{main}$ and $Loss_{aux}$ are cross-entropy losses.

2.4. Our Proposed Network

We propose a new network, UCU-Net, with the encoder-decoder architecture, for vessel segmentation. Fig. 2 shows UCU-Net, which is essentially a combination of U-Net [10] and CUMed-Vision Network [11]. UCU-Net has the same encoder as U-Net; in its decoder, side outputs from the last stage of the encoder and from all the stages of the decoder (shown with * in Fig. 2) are combined to generate the final output of the network. Such combined decoder facilitates multi-stage contextual information flow, achieving enhanced segmentation compared to U-Net and CUMed-Vision Network, which we found during initial experiments. By merging the outputs of different decoder stages in the $Loss_{main}$ calculation, while allowing information flow among different decoder layers, it enables the network to learn improved features.

3. EXPERIMENTS

In Sec. 3.1, we discuss the datasets used to verify our framework. Sec. 3.2 explains the experimental setup. We provide

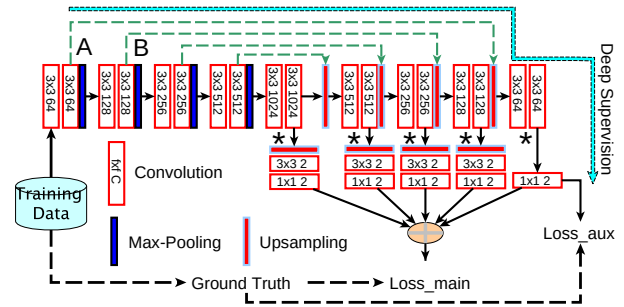


Fig. 2. Our UCU-Net architecture. $f \times f$ is the filter dimension and C is the number of filters in each convolutional layer.

the details of the results obtained and the ablation study performed, in Sec. 3.3 and Sec. 3.4, respectively.

3.1. Datasets

Three publicly available retinal vessel segmentation datasets, DRIVE [13], CHASE_DB1 [1], and STARE [14], are used to evaluate our method. The DRIVE dataset consists of 20 training and 20 testing retinal images, where each of these images is of size 584×565 . The CHASE_DB1 dataset contains 28 retinal images of size 999×960 . STARE contains 20 retinal images of size 700×605 . There are no specifically allocated training and test images for these two datasets. Following [5], for CHASE_DB1, we use the first 20 images as training images and the last 8 images for testing. For STARE, we use the leave-one-out approach following [3] for experiments.

3.2. Experimental Setup

We resize the images (DRIVE: 512×512 , CHASE_DB1: 976×976 , STARE: 656×656), and preprocess them as in [5]. To extract $LERF$ from RF , a threshold of 2 standard deviations (95.45%) of the *Gaussian* distribution of RF , is used [9]. Mean of 20 iterations of the square root of $LERF$ region is considered as the size of $LERF$ in order to cancel out the variations caused by the *ReLU* layers. The average vessel widths for resized images are found to be 3.33 (DRIVE), 6.16 (CHASE_DB1), and 4.25 (STARE), and all are best represented by the 2^{nd} convolutional layer of our network (A in

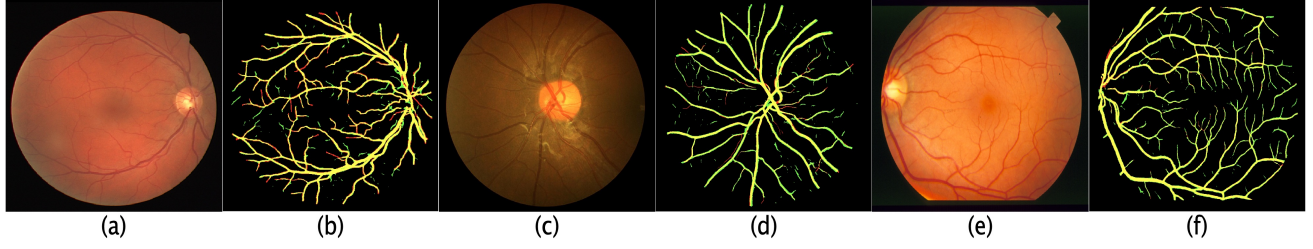


Fig. 3. Sample test images from (a) DRIVE, (c) CHASE_DB1, and (e) STARE datasets. (b), (d), and (e) Vessels corresponding to the ground truth and segmentation outputs are shown in Red and Green channels, respectively (Yellow = Red + Green).

Table 1. Accuracy comparison on DRIVE, CHASE_DB1, and STARE datasets.

Method	DRIVE [13]				CHASE_DB1 [1]				STARE [14]			
	AUC (%)	Acc (%)	Sp (%)	Se (%)	AUC (%)	Acc (%)	Sp (%)	Se (%)	AUC (%)	Acc (%)	Sp (%)	Se (%)
Fraz et al. [1]	97.47	94.80	98.07	74.06	97.12	94.68	97.11	72.24	97.68	95.34	97.63	75.48
Li et al. [2]	97.38	95.27	98.16	75.69	97.16	95.81	97.93	75.07	98.79	96.28	98.44	77.26
Orlando et al. [3]	95.07	-	96.84	78.97	95.24	-	97.12	72.77	-	-	97.38	76.80
Zhang et al. [4]	97.99	95.04	96.18	87.23	99.00	97.70	99.09	76.70	98.82	97.12	99.01	76.73
Xia et al. [5]	98.07	95.67	98.19	78.44	98.25	96.37	98.47	75.38	-	-	-	-
Proposed	97.24	95.40	96.01	89.16	97.63	96.01	96.51	88.05	97.42	95.71	96.34	87.71

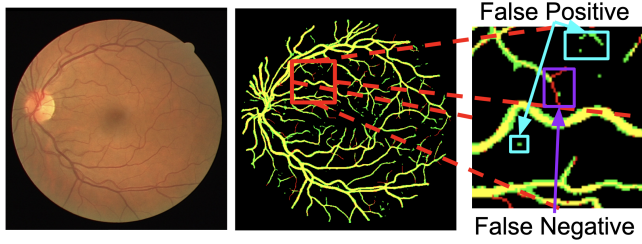


Fig. 4. (Left) A test image from the DRIVE dataset. (Middle) Ground truth and segmentation output are shown in Red and Green channels (Yellow = Red + Green). (Right) Highlighted section showing some false positive vessels.

Fig. 2). As in [10], random flipping, rotation, and elastic deformations are used for data augmentation. We use a standard back-propagation implementing *SGD* with the max iteration number equal to $2k$. Initial learning rate is set to 0.005 and is halved in every 200 epochs. Class balanced weights for loss calculation are adopted due to the imbalance in class distribution. Experiments are performed on NVIDIA-TITAN and Tesla P100 GPUs, using the Torch framework with *He* initialization [15]. We use a similar setup for training as provided in [10]. However, we use a padded U-Net removing the requirement of cropping in later stages. Without any post-processing, the network output is used for evaluation.

3.3. Results

Segmentation results on the DRIVE, CHASE_DB1, and STARE datasets are summarized in Table 1. Sample outputs for all 3 datasets are provided in Fig. 3. It is evident that our approach achieves significant increase in sensitivity (Se) in all 3 cases by reducing the amount of false negative vessels. However, existence of additional false positive

Table 2. Ablation study on the DRIVE dataset.

Method	AUC (%)	Acc (%)	Sp (%)	Se (%)
M+A	97.24	95.40	96.01	89.16
M	96.71	96.08	97.10	85.32
M+B	96.93	95.35	95.99	88.69

vessels causes decrease in specificity (Sp) (Fig. 4). Direct inclusion of features extracted in the initial layers (because of thin vessels) makes the network highly sensitive. This results in false positives which can be removed using simple connected-component based post-processing.

3.4. Ablation Study

The DRIVE dataset is used as an example case for ablation study. The effect of data-aware deep supervision is shown in Table 2. The *M+A* case shows the accuracy obtained with both the main, and auxiliary losses determined using our proposed framework (from A in Fig. 2), while *M* shows the accuracy obtained with only the main loss. For *M+B*, in addition to the main loss, deep supervision is used from location B in Fig. 2. In the absence of deep supervision, the network compromises sensitivity while achieving better specificity.

4. CONCLUSIONS

In this paper, we explored data-aware deep supervision for retinal vessel segmentation. For enhancing thin vessel segmentation, we introduced deep supervision utilizing the labeled training data and *LERF*. We validated our approach on the DRIVE, CHASE_DB1, and STARE datasets on a new network architecture (i.e., UCU-Net). Our framework significantly increases sensitivity over the state-of-the-art, while maintaining comparable AUC, accuracy, and specificity.

5. REFERENCES

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